

Supplementary Material for PACE-Seg: When Does Input-Adaptive Semantic Segmentation Pay Off on Edge Accelerators?

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Abstract

This supplement records checkpoint provenance, the measurement protocol, Hailo-8 sensitivity configurations, compiled-engine accuracy, calibration quality, condition-level routing results, and additional qualitative examples for the main manuscript. All numbers come from the same landscape-input training runs (seeds 0–2) and measurements as the manuscript; no new claims or evaluation samples are introduced.

1. Evidence scope

The main manuscript reports three findings: a FLOPs-proportional proxy does not reproduce measured feasibility; weight sharing costs the largest capacity 0.9–2.8 mIoU points relative to training it alone; and, once routes and static deployments are timed in the same harness on the two primary backends, routing under a hard budget beats rank escalation but not static deployment; in a counterfactual overhead-scaling analysis, the static advantage would vanish under mean-cost budgets only below about 1.3 ms (AGX Xavier) and 4.3 ms (Hailo-8) of overhead. This supplement contains the material needed to reproduce and audit these results. Operating cells are correlated evaluations, and the two devices share segmentation predictions. Two budget grids are used: the route-budget grid ($3 \text{ runs} \times 2 \text{ backends} \times 5$

splits $\times$ 4 budgets) and, for the comparison with static deployment, a static-own-cost grid per backend (3 runs $\times$ 5 splits $\times$ 8 budgets), part of which lies below the cheapest route.

2. Training and checkpoint provenance

All models are trained at the native 2:1 aspect ratio of Cityscapes and ACDC, with input sizes 384×192 , 512×256 , 768×384 , and 1024×512 (width $\times$ height) for tiny to large. Elastic training uses the sandwich rule with boundary-aware cross-entropy and in-place distillation from the largest level ($\alpha = 0.5$, $T = 1$); 100,000 steps, batch size 8, AdamW (learning rate 3×10^{-4} , weight decay 10^{-4}), 500 warm-up steps with cosine decay, gradient clipping at 5.0, and FP16 mixed precision. The references for RQ2, Fast-SCNN and the large architecture trained alone (PACE-Large), use the same recipe and three seeds each. Table S1 lists the supernet checkpoints used by every router analysis.

Table S1: Supernet checkpoints used by the router analyses (Runs A–C = seeds 0–2). Hashes are abbreviated; full SHA-256 values are stored in the per-image result files.

Run	Seed	Step	Checkpoint SHA-256	Config hash
A	0	100,000	f19833d5	7d6c05422dbe
B	1	100,000	90978478	a7946574eb05
C	2	100,000	53516de6	e26a2c8985bc

3. Hardware and latency protocol

Table S2 lists the toolchains. TensorRT engines are FP16. Hailo-8 HEFs are INT8, compiled from the same fixed-shape ONNX graphs with

the Dataflow Compiler and quantized with 200 real fit-half images. The Jetson DLA is excluded because decoder layers fall back to the GPU.

Table S2: Hardware and compiler stacks.

Name	Hardware	Runtime and system versions
Hailo-8	Raspberry Pi 5 + Hailo-8 M.2	HailoRT 4.23.0; Dataflow Compiler 3.34.0; Model Zoo 2.19.0
Xavier NX	Jetson Xavier NX	L4T R35.4.1; CUDA 11.4.19; cuDNN 8.6.0.166; TensorRT 8.5.2.2
AGX Xavier	Jetson AGX Xavier	L4T R35.6.4; CUDA 11.4.19; TensorRT 8.5.2.2
Orin Nano	Jetson Orin Nano Super	L4T R36.5.0; CUDA 12.6.68; cuDNN 9.3.0; TensorRT 10.3.0.30

Candidate-only latency (RQ1) uses the trtexec/hailortcli protocol at batch size one on all four devices. Complete-route cost $C_h(\ell)$ and static cost $S_h(\ell)$ are measured on AGX Xavier and Hailo-8 in one process, with the same engines, inputs, and timer boundary: the timer starts when a preprocessed host tensor is handed to the runtime and stops when the returned prediction is synchronized; decoding, resizing, and normalization are excluded. A route runs the tiny engine, entropy, and the policy decision and, unless tiny is selected, one further engine. Each configuration uses 50 warm-up and 500 timed iterations; the median, p95, and p99 are recorded. Cold-start costs are excluded.

Figure S1 compares candidate-only p95 latency with the complete-route median. On Hailo-8 the short tiny route expands by $11.0\times$ because network-

group activation and host-side entropy dominate it.

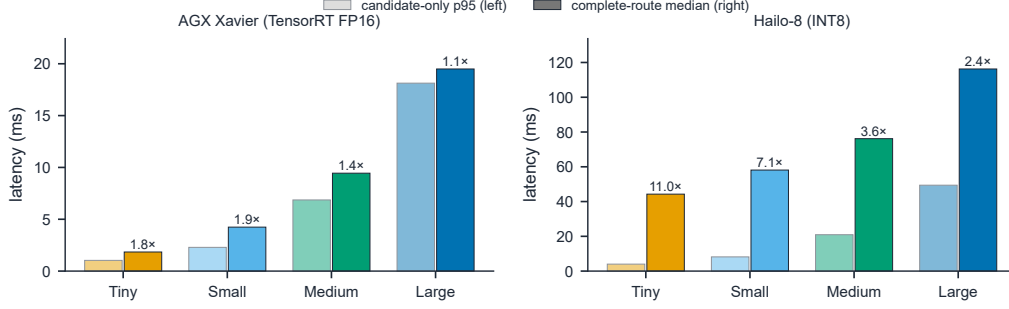


Figure S1: Candidate-only p95 latency (left bars) versus complete-route median (right bars); ratios above the bars give the expansion. Bars compare two measured totals and are not a component decomposition.

Hailo-8 configuration sensitivity. The primary Hailo-8 path uses explicit network-group activation with FLOAT32 output streams. Table S3 repeats the same-harness comparison with the HailoRT model scheduler and with UINT8 output streams. The routing overhead stays between 36 and 50 ms in every configuration. In each configuration 45 of the 120 static-own-cost cells lie below the cheapest route, where only static deployment can meet the budget; on the remaining route-feasible cells static deployment beats routing in every configuration.

Table S3: Same-harness routing overhead and policy-D comparisons per device configuration (median costs, mIoU points). Feasible: route-feasible cells of the static-own-cost grid, with paired image-level 95% bootstrap CI (1,000 replicates, image sampling only) and cell count; all (diag.): all 120 cells, including route-infeasible budgets where D violates; route-infeasible cells: cells where only static deployment fits. Mixture gain: candidate-specific routing without a per-frame cap minus the static mixture at equal mean cost.

Configuration	Overhead $C - S$ (ms)	D – static, feasible [95% CI] (n)	D – static, all (diag.)	Route-infeasible cells	Mixture gain
AGX Xavier (logits)	0.7–2.7	-3.00 [-3.07, -2.72] (105)	-2.62	15/120	-0.52
Hailo-8 explicit, FLOAT32 (primary)	38.0–46.9	-8.18 [-8.42, -7.40] (75)	-7.41	45/120	-5.52
Hailo-8 explicit, UINT8	42.6–49.7	-10.97 [-11.26, -9.94] (75)	-9.15	45/120	-6.33
Hailo-8 scheduler, FLOAT32	36.4–49.3	-8.18 [-8.42, -7.40] (75)	-7.41	45/120	-5.50
Hailo-8 scheduler, UINT8	41.2–50.0	-10.97 [-11.26, -9.94] (75)	-9.15	45/120	-6.33

4. Compiled-engine accuracy

Table S4 compares compiled engines of Run A with the PyTorch predictions used in the replay, on the held-out halves. TensorRT FP16 is within 0.02 points everywhere. Hailo-8 INT8 loses up to 1.6 points on tiny and at most 0.5 points on large; the probe entropy keeps a rank correlation of at least 0.88 with the PyTorch probe.

Table S4: Compiled minus PyTorch mIoU (points) per capacity and split, and the minimum pixel agreement over splits (Run A, held-out halves).

Backend	Capacity	Cityscapes	Fog	Night	Rain	Snow	Min. agreement (%)
AGX Xavier, TensorRT FP16	tiny	-0.01	-0.00	-0.01	-0.01	-0.00	99.9
AGX Xavier, TensorRT FP16	small	-0.00	+0.01	+0.01	-0.01	+0.00	99.7
AGX Xavier, TensorRT FP16	medium	-0.01	+0.00	-0.01	+0.00	+0.00	99.8
AGX Xavier, TensorRT FP16	large	-0.02	-0.01	+0.01	-0.01	-0.01	99.8
Hailo-8, INT8	tiny	-1.64	-0.96	-1.62	-0.59	-0.51	87.3
Hailo-8, INT8	small	-0.52	-0.32	-0.33	-0.31	+1.87	93.6
Hailo-8, INT8	medium	-0.41	-0.28	-0.43	+0.21	-0.30	95.7
Hailo-8, INT8	large	-0.09	-0.36	-0.52	-0.02	-0.13	95.8

5. Router protocol and leakage safeguards

Segmentation quality is dataset-level mIoU, computed once from the accumulated confusion matrices of the selected candidates. Each validation split is divided by alternating index into fit and held-out halves. Calibrators, risk targets, T-hard thresholds, and operating points use only fit halves; held-out labels are read only after selection. Five risk targets are the 10th–90th percentiles of fit-half observed error, macro-aggregated across splits, with one grid per run shared by both devices. A cell is violating if at least one held-out image is routed to a candidate whose tabulated cost exceeds the budget; fair cells for comparisons with policy A are those in which A has no violation.

6. Calibration quality

Table S5 reports held-out calibration quality per split. Figure S2 shows reliability pooled over all splits. All four calibrators lie close to the diagonal; replacing ten equal-count bins by five bins or isotonic regression changes the held-out error by at most 0.003, whereas twenty bins overfit the small ACDC fit halves. Calibration quality is therefore not the reason that candidate-specific calibration adds nothing over T-hard in the manuscript.

Table S5: Held-out calibration quality. MAE of the configured calibrators (tiny/small/medium/large); ρ_s : Spearman correlation between probe score and observed error (tiny/large); MAE averaged over candidates for 5/10/20 equal-count bins and isotonic regression; Spearman correlation between per-image pixel error and per-image 1–mIoU.

Split	MAE (T/S/M/L)	ρ_s (T/L)	5 / 10 / 20 bins	Isotonic	Error vs 1–mIoU
Cityscapes	.028/.025/.021/.019	.81/.74	.023/.023/.023	.022	.50
ACDC fog	.031/.027/.022/.019	.68/.61	.026/.025/.033	.024	.23
ACDC night	.051/.055/.057/.054	.67/.37	.054/.054/.059	.053	.58
ACDC rain	.032/.032/.030/.024	.67/.61	.029/.030/.032	.028	.32
ACDC snow	.045/.038/.034/.029	.71/.58	.035/.036/.046	.034	.48

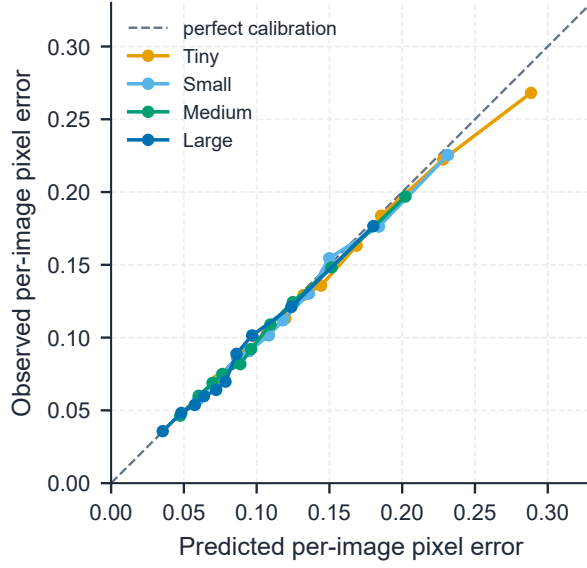


Figure S2: Held-out reliability of the four candidate-specific calibrators, pooled over Runs A–C and the five splits.

7. Condition-level routing and LOCO

Table S6 disaggregates the comparison with policy A by condition (24 cells per row). In the leave-one-condition-out analysis, calibrators, risk-target grid, and operating points are refitted without the evaluated condition; D then has no violating cell, beats A in all 29 decided fair cells (12 ties, +4.15 points), and beats A-hard by +1.59 points over all cells (48/72/0). Mean LOCO D quality is 45.2, 43.1, 28.7, 40.0, and 39.1 mIoU on Cityscapes, fog, night, rain, and snow.

Table S6: Condition-level configured and pooled D against policy A (median costs). W/T/L and mean Δ use fair cells; violation counts use all 24 cells per condition.

Condition	Fair	Config. W/T/L	Config. Δ	Pooled W/T/L	Pooled Δ	A viol.	D viol.
Cityscapes	6/24	6/0/0	+8.81	6/0/0	+8.87	18/24	0/24
ACDC/Fog	9/24	9/0/0	+5.38	9/0/0	+5.51	15/24	0/24
ACDC/Night	14/24	14/0/0	+1.05	14/0/0	+1.08	10/24	0/24
ACDC/Rain	21/24	15/6/0	+2.09	15/6/0	+2.11	3/24	0/24
ACDC/Snow	12/24	10/0/2	+1.49	10/0/2	+1.48	12/24	0/24

8. Robustness analyses

These analyses replay stored predictions and measured cost tables; no model was retrained and no device was re-measured. They are robustness checks and do not replace the primary results.

Per training seed and sequence-grouped split. Table S7 repeats the main comparisons for each training run separately and for a fit/held-out split by whole recording (ACDC) or city sequence (Cityscapes) instead of alternating index; calibrators, risk-target grid, T-hard thresholds, and operating points are refitted on the new fit half only. Night and fog contain only two and three recordings, so their sequence split places different recordings in the two halves. Every conclusion holds in every seed and under the sequence split; the sign of D – T-hard changes across seeds, consistent with no detectable difference.

Table S7: Mean mIoU difference (points) with the number of cells in parentheses, per training run (seeds 0–2), pooled (as in the manuscript), and with a sequence-grouped fit/held-out split. Static rows use route-feasible cells of the static-own-cost grid.

Comparison (median)	Run A	Run B	Run C	Pooled	Sequence split
D – A, fair	+2.70 (21)	+3.13 (23)	+2.74 (18)	+2.87 (62)	+3.38 (55)
D – A-hard	+1.33 (40)	+1.56 (40)	+1.35 (40)	+1.41 (120)	+1.71 (120)
D – T-hard	-0.11 (40)	-0.06 (40)	+0.09 (40)	-0.03 (120)	-0.04 (120)
D – static, route cost	-0.19 (40)	-0.18 (40)	-0.13 (40)	-0.16 (120)	-0.20 (120)
D – static own, AGX, feasible	-3.05 (35)	-3.00 (35)	-2.94 (35)	-3.00 (105)	-2.98 (105)
D – static own, Hailo-8, feasible	-8.42 (25)	-8.18 (25)	-7.95 (25)	-8.18 (75)	-8.39 (75)

Break-even interval. An image-level bootstrap (200 replicates, fixed seed) of the counterfactual mean-cost break-even overhead gives 1.27 ms [0.57, 1.50] on AGX Xavier and 4.26 ms [2.13, 5.15] on Hailo-8, against measured overheads of 1.9 ms and 44 ms; every replicate crossed zero. The interval covers image sampling only; the overhead grid is coarse, and seed and timing variability are not included.

9. Additional qualitative examples

All examples are held-out images of Run A, fixed by rules applied to the recorded per-image errors before visual inspection: per split, the image with the largest tiny-minus-large pixel error (Fig. S3), the image nearest the median large-candidate error (Fig. S4), the image with the largest large-candidate error (Fig. S5), and, among images with below-median tiny error, the one with the smallest tiny-minus-large difference (Fig. S6). The routed output is policy D on AGX Xavier at the budget of the large route. Dashed boxes mark the window where tiny and large disagree most; insets enlarge it; orange in the error column marks disagreement with ground truth.

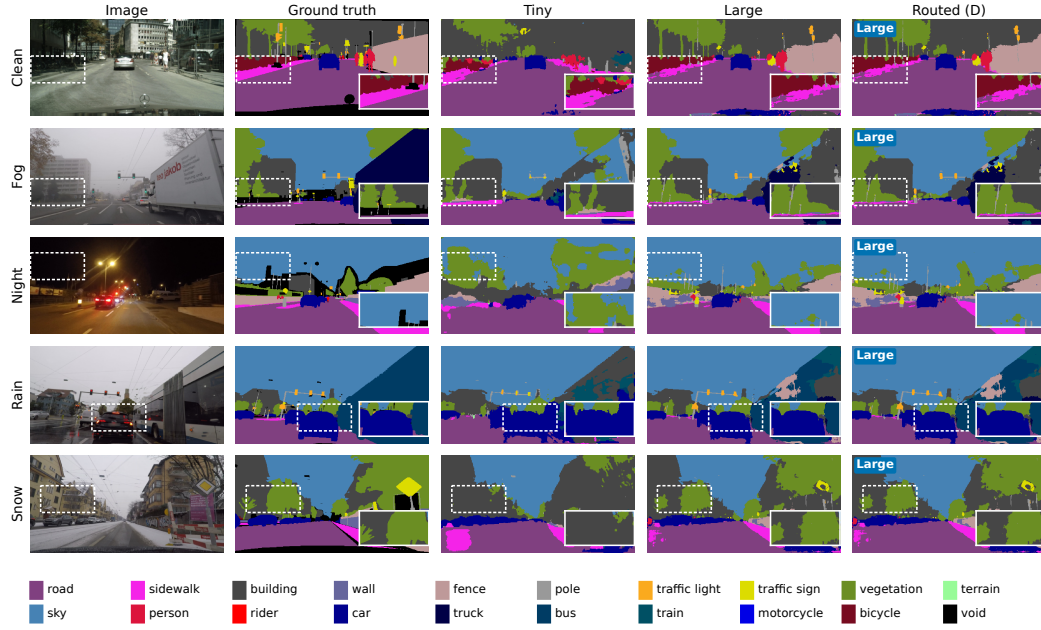


Figure S3: Largest tiny-minus-large error per condition: where capacity matters most.

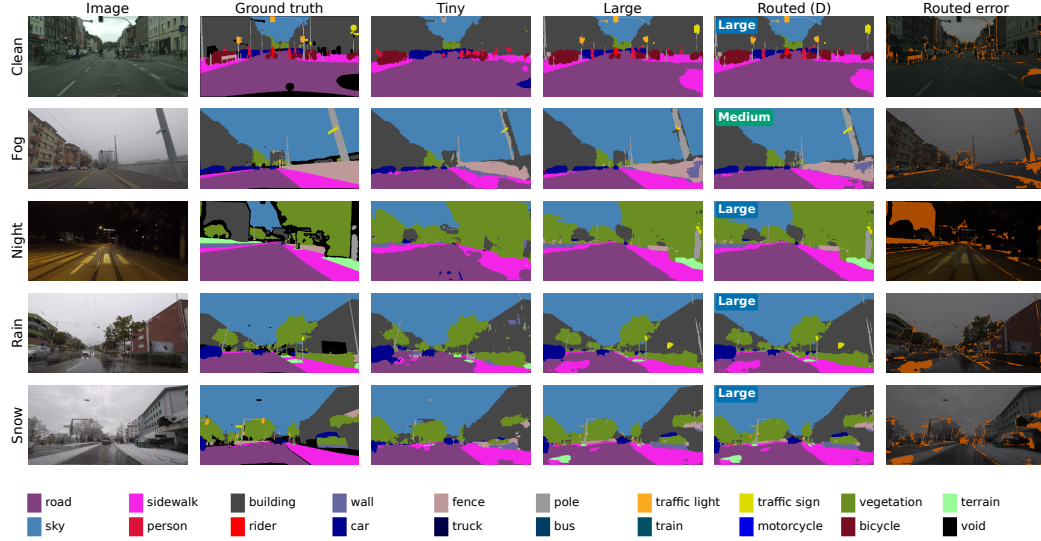


Figure S4: Median-error examples per condition, with the routed output and its error map.

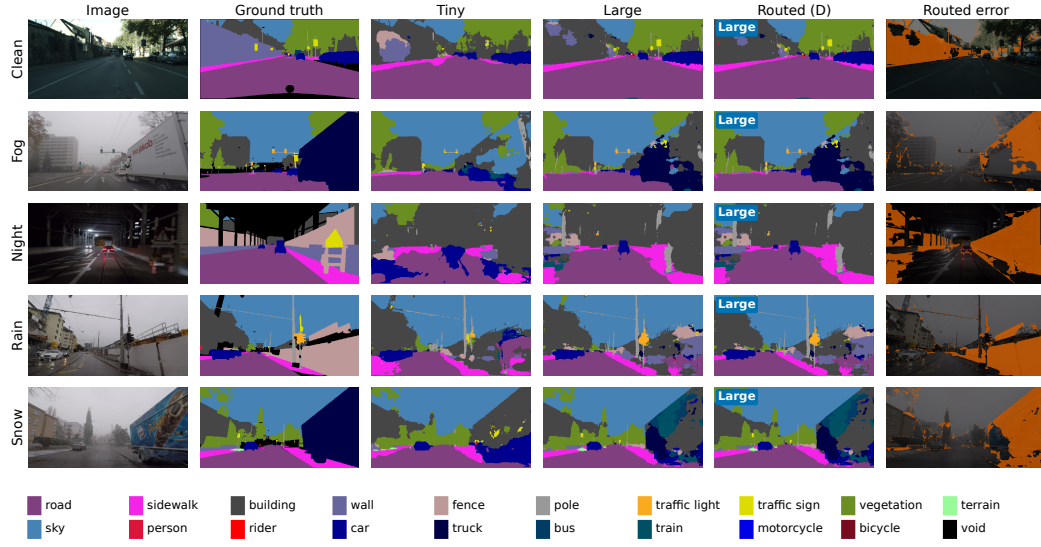


Figure S5: Hardest example per condition (largest large-candidate pixel error), illustrating remaining failure modes; not an estimate of a failure rate.

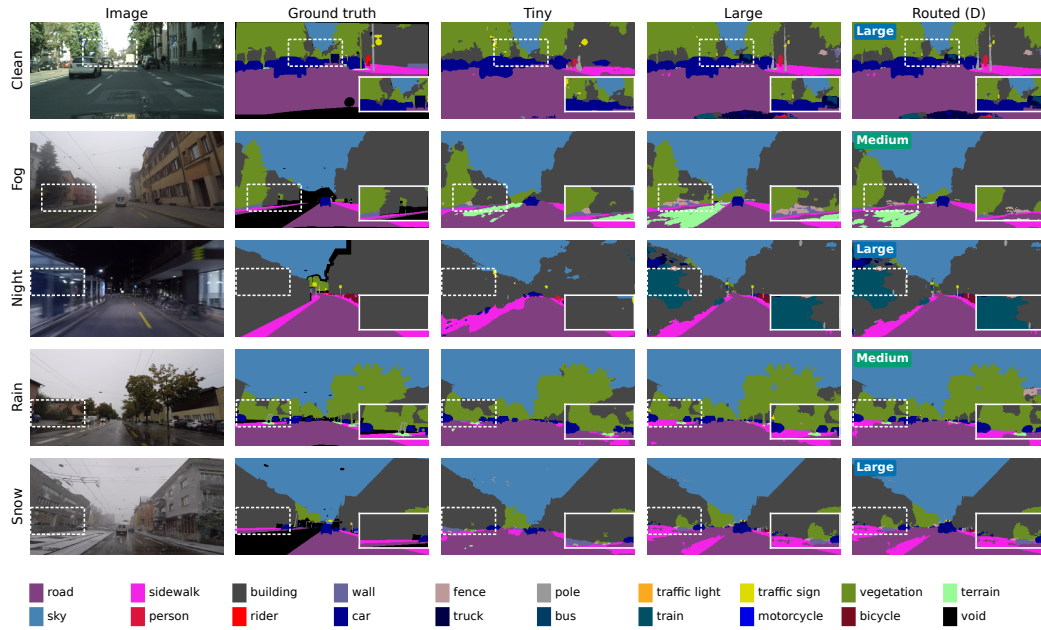


Figure S6: Examples on which tiny is already close to large.

10. Evidence locations

Per-image router dumps are in `reports/landscape_20261004/`; same-harness latency in `reports/static_vs_route*_20261004.json`; router analyses in `reports/router_review_analyses_20261004.json`, `router_same_harness_analysis_20261004.json`, `router_breakeven_20261004.json`, `router_nested_bootstrap_20261004.json`, and `router_by_condition_20261004.json`; compiled-engine accuracy in `reports/compiled_eval_E*_20261004.json`; qualitative assets with their selection rules in `reports/qualitative_assets_v20*/`. Cityscapes and ACDC data and training checkpoints are not redistributed.